Deep Learning Laboratory - Assignment 5

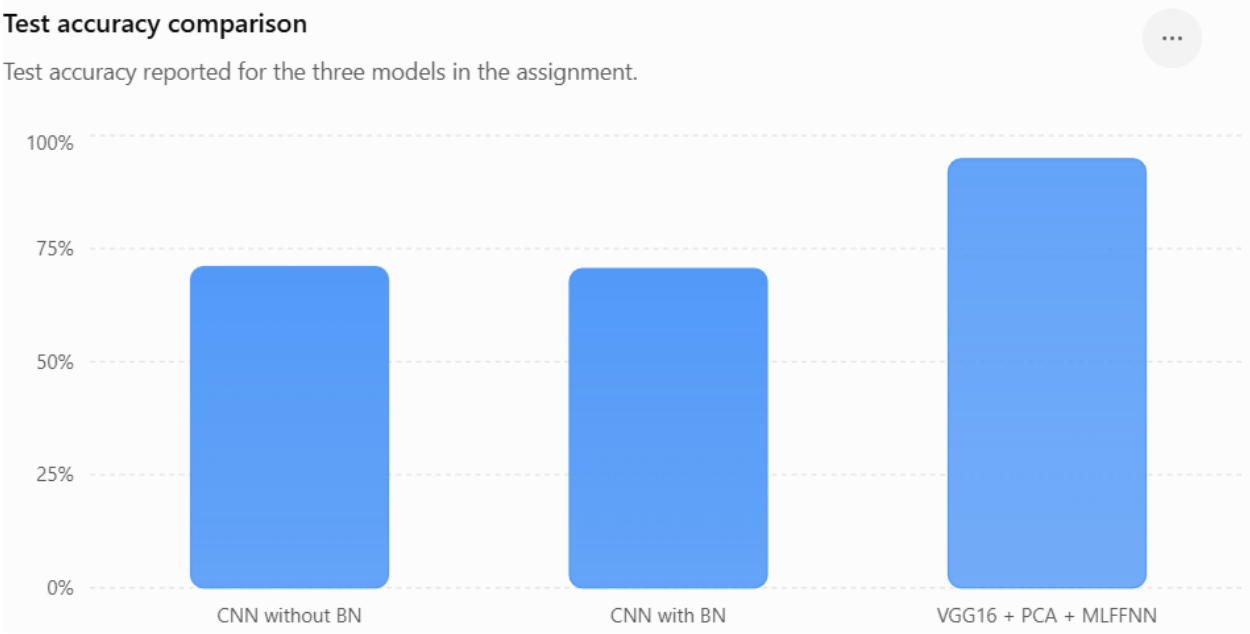
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Setup

Data	5 classes, 96×96×3; 3501 train / 700 test
CNN	Conv(3×3, S=1, P=1, 4) → ReLU → [BN] → MaxPool(2×2, S=2) Conv(3×3, S=1, P=1, 6) → ReLU → [BN] → MaxPool(2×2, S=2) → FC(3456→5)
Transfer	VGG16 (frozen) → 4096-D → PCA (95%, 984-D) → 256 → 64 → 5; Tanh
Output / loss	Softmax output; cross-entropy loss

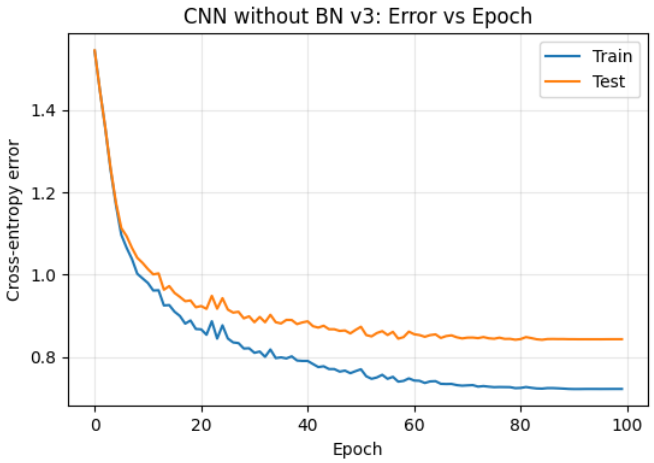
Accuracy

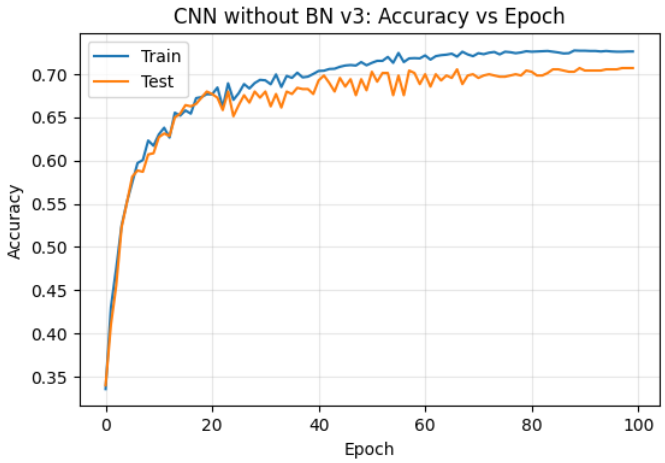
Model	Seed 42 (%)	Seed 0 (%)	Seed 1 (%)	Test (%)
CNN without BN	70.71	68.43	73.86	71.00 ± 2.23
CNN with post-activation BN	68.29	70.43	73.00	70.57 ± 1.93
VGG16 + PCA + MLFFNN		single run		**94.86**

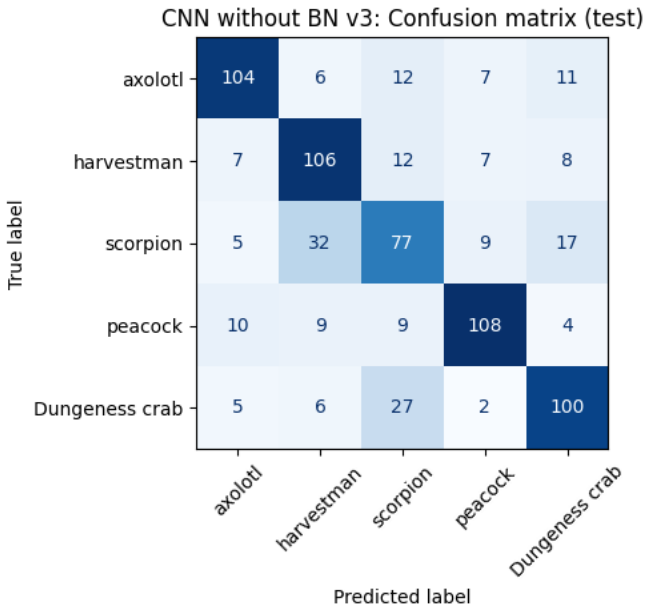


Error curves and confusion matrices

CNN without Batch Normalization (seed 42)



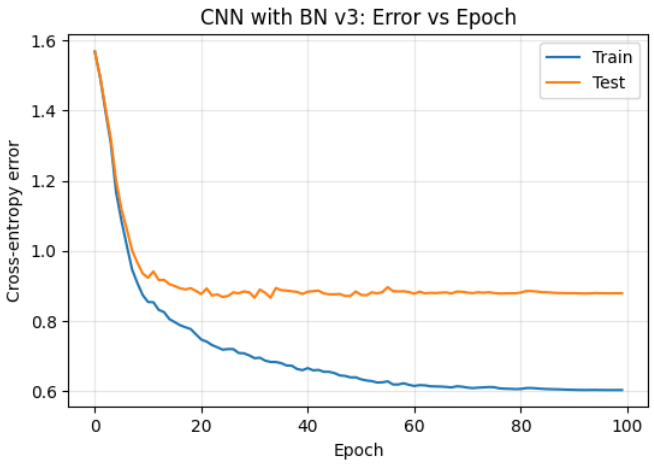


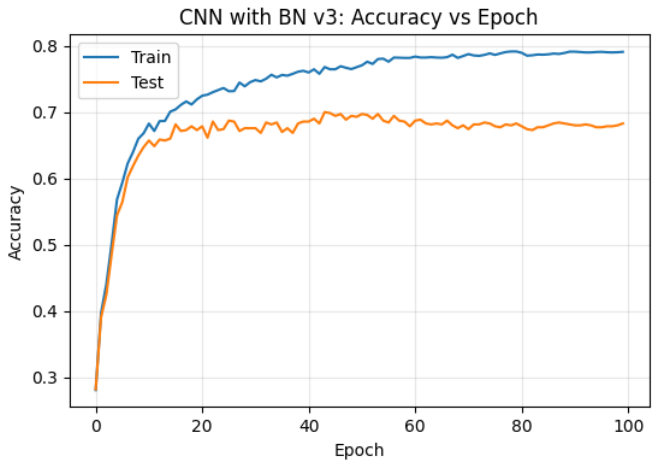


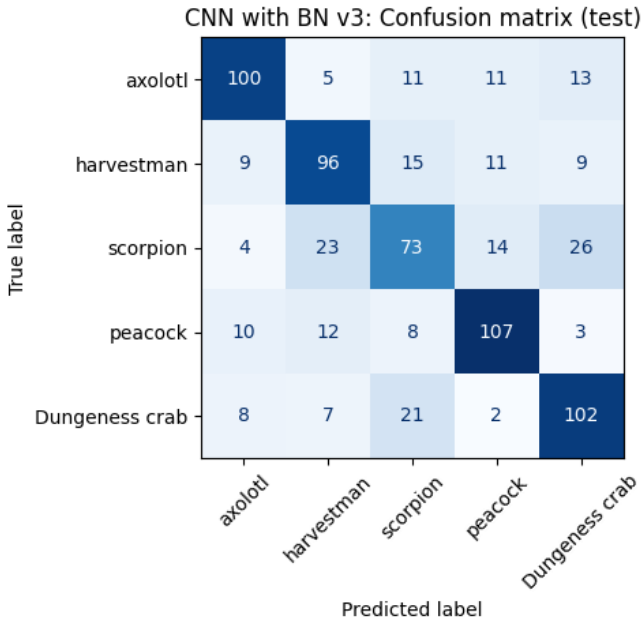
Observations

- Training accuracy reaches around 72–73%.
- Test accuracy reaches around 71%.
- Training and test curves remain relatively close.
- Test loss settles around 0.84.

CNN with Batch Normalization (seed 42)



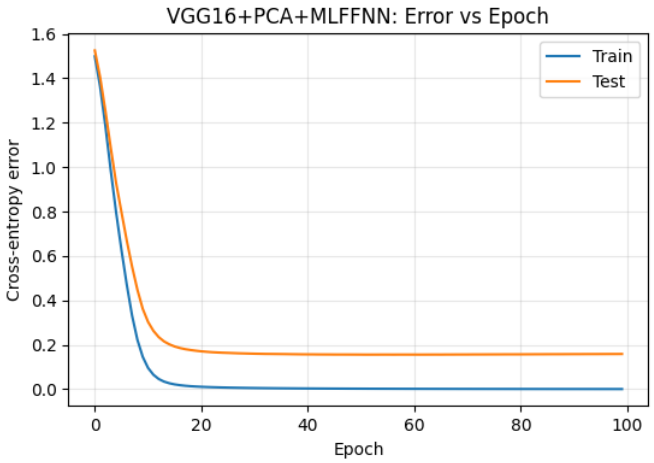


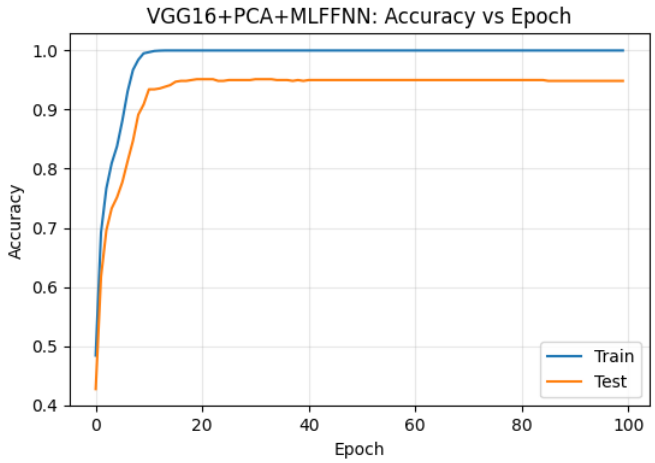


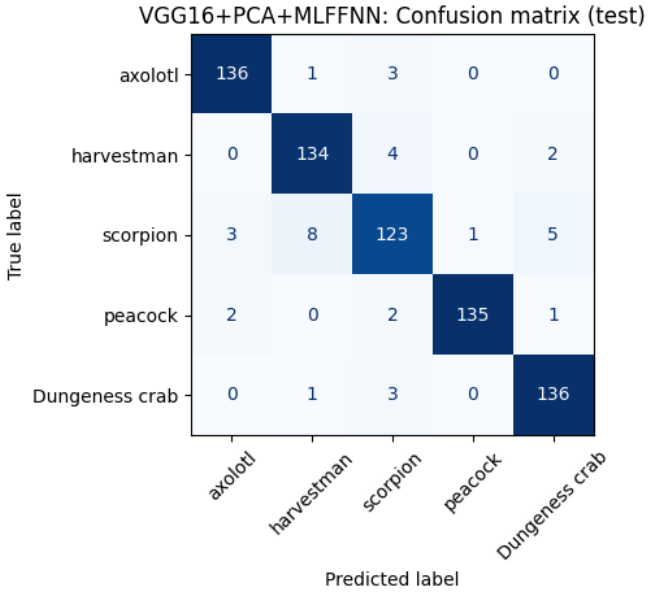
Observations

- Training accuracy reaches around 79%.
- Test accuracy settles around 68–69% in the displayed seed-42 plot.
- There is a noticeable train–test gap.
- Test loss remains around 0.88.

VGG16 + PCA + MLFFNN







Observations

- Training accuracy reaches approximately 100%.
- Test accuracy reaches approximately 95%.
- Test loss decreases to approximately 0.16.
- The confusion matrix shows substantially stronger classification performance than the two small CNNs.

CNN settings

Setting	Value
Input image size	96×96×3
Number of classes	5
Convolution / pooling layers	2 / 2
CL1 / CL2 feature maps	4 / 6
Convolution filter, stride, padding	3×3, 1, 1
Activation	ReLU
Pooling	Max, 2×2, stride 2
Optimizer	Adam
Epochs	100
Mini-batches per epoch	5
Data augmentation	Random horizontal flip + small random shift
Learning-rate schedule	Cosine annealing
Loss	Cross-entropy
Random seeds	42, 0, 1

CNN-specific hyperparameters

Hyperparameter	CNN without BN	CNN with BN
Learning rate	0.002	0.001
Weight decay	1×10^{-5}	5×10^{-4}
Optimizer	Adam	Adam
Scheduler	Cosine annealing	Cosine annealing
Epochs / mini-batches	100 / 5	100 / 5
Augmentation	Yes	Yes

VGG16 + PCA + MLFFNN settings

Setting	Value
Backbone	Pretrained VGG16 (ImageNet), frozen
Input to VGG16	224×224
Representation	Pre-final FC layer, 4096-D
PCA	95% variance (984-D), fitted on training data only
Standardization	StandardScaler
Hidden layer 1	256 neurons, Tanh
Hidden layer 2	64 neurons, Tanh
Output	5 neurons, softmax
Optimizer / learning rate	Adam / 0.001
Epochs / mini-batches	100 / 5
Loss	Cross-entropy

Results

The CNN without Batch Normalization achieved a test accuracy of 71.00%, while the CNN with Batch Normalization achieved 70.57%. The VGG16 + PCA + MLFFNN model achieved a test accuracy of 94.86%, which is substantially higher than both CNN models. Overall, the results show that the VGG16-based transfer learning approach combined with PCA and MLFFNN achieved the highest test accuracy among the three models.